AP-E6 Same-Soliton Yukawa–Callias Audit: The Physical Moduli No-Go, a Constituent Dirac Completion, CPT Algebra, and a Typed Local Mixed-WZW Ansatz

Route F research note

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Abstract

We test whether the canonical Workline-B relaxed charged two-colour $B = 1$ field can support the same physical $\mathbb{CP}^1$ fermion family required by AP-E4. The answer is a sharp no-go for the present model and a correctly typed local differential-character ansatz; neither is a physics-promotion claim. The background is bound to the Workline-B public solver and machine card by the SHA-256 checksum of the canonical little-endian float64 (r, F, s) array.

First, the internal stabilizer of

$$U_F(\boldsymbol{x}) = \cos F(r) + \mathrm{i}\,\boldsymbol{\tau}\cdot\widehat{\boldsymbol{x}}\sin F(r)$$

is only the centre $\mathbb{Z}_2$. Its pure isospin orbit is therefore $SU(2)/\mathbb{Z}_2 \simeq SO(3)$, not $\mathbb{CP}^1$. The same $SO(3)$ is obtained from the combined spatial–isospin symmetry after quotienting the diagonal hedgehog stabilizer; these are not two independent collective-coordinate sets and neither is a colour-gauge redundancy. The separate scalar-vacuum $\mathbb{CP}^1$ is spatially uniform and has norm proportional to the three-volume, so it is not a localized normalizable soliton modulus.

Second, we write the minimal declared constituent chiral Yukawa completion and its static Hamiltonian,

$$\mathcal{D}_E = \gamma^\mu D_\mu + M(P_R U_F + P_L U_F^\dagger), \qquad H_F = -\mathrm{i}\alpha^i D_i + \beta M\big(\cos F + \mathrm{i}\gamma_5\boldsymbol{\tau}\cdot\widehat{\boldsymbol{x}}\sin F\big).$$

Writing the Hermitian potential as $\Phi_F = \beta M(\cos F + \mathrm{i}\gamma_5\boldsymbol{\tau}\cdot\widehat{\boldsymbol{x}}\sin F)$, its asymptotic mass is $\Phi_\infty = \beta M$, not $M\mathbf{1}$. Under the stated domain, bounded-commutator, coercivity, and decay hypotheses, the ordinary S^2_∞ Callias positive-eigenbundle is constant and the static Callias index is zero. The four-dimensional interpolation formula $\mathrm{Ind}(\partial_\tau + H_\tau) = \mathrm{SF}(H_\tau) = B$ concerns spectral flow and does not force a zero mode at the final static soliton. A finite Wilson control finds no accidental endpoint zero at the benchmark, but does not prove a continuum spectral gap.

Third, a new two-band obstruction rules out the previously declared mixed eigenline in the minimal mass representation. On $X = S^2_\infty \times \mathbb{CP}^1$, let x, y generate the two degree-two cohomologies. A globally nondegenerate 2×2 Hermitian mass with one positive band defines a map $f : X \to \mathbb{CP}^1$; its eigenline obeys $c_1^2 = f^*(u^2) = 0$. But the desired line has

$$c_1 = x + 2y, \qquad c_1^2 = 4xy \neq 0.$$

Thus a nontrivial ambient bundle, at least three bands, or another microscopic mechanism is necessary. The numerical $\mathcal{O}(2)$ Veronese line is retained only as a spectator control.

Finally, define first the universal differential character

$$\widehat{\Xi}_5 = 2\,\mathrm{pr}_U^*\,\widehat{\omega}_3 \smile \mathrm{pr}_s^*\,\widehat{c}_1(\mathsf{H}) \in \widehat{H}^5(SU(2)\times\mathbb{CP}^1;\mathbb{Z}),$$

then pull it back by the evaluation map on $\Sigma_3 \times \mathcal{C}_{\mathrm{restr}}$ and fiber-integrate to a degree-two character. This is a well-typed local ansatz. Constant-loop and centre phase controls do not prove descent on the restricted gauge stack: all components of $\mathrm{Map}(\Sigma_3, U(1))$, an equivariant cocycle, and vertical holonomy remain untreated. The odd/even centre test is only a necessary control, not an exactly-two theorem. Consequently every descent, same-soliton composition, and portal gate remains false.

Contents

1 Question, evidence levels, and theorem-level answer

The previous AP-E4 note supplied a conditional families-Callias template. The present task is stronger: start from the *same* nonlinear $B = 1$ configuration published by charged two-colour Workline-B, determine its true symmetry orbit, and only then ask for a fermion family, determinant line, CPT map, and WZW line. We use four evidence labels.

Same-background result. A statement computed from the canonical Workline-B $(F(r), s(r))$ and the fields already present in the charged EFT.

Declared completion. An explicit new fermion term that is compatible with the displayed low-energy symmetries but has not been matched from the microscopic charged gauge theory.

Restricted theorem. A complete construction on a stated open subspace, with every required equivariance and globality hypothesis stated.

Spectator control. A finite matrix or bundle designed to validate an index, Berry, or regulator formula but not derived from the mother model.

Theorem 1.1 (Strongest defensible AP-E6 result)**.** *For the present $B = 1$ hedgehog, the localized rotational/isospin orbit is $SO(3)$, while the scalar $\mathbb{CP}^1$ vacuum orientation is nonnormalizable in infinite volume. Hence there is no same-soliton physical $\mathbb{CP}^1$ on which an $\mathcal{O}(2)$ families determinant line or its $\mathcal{O}(-4)$ CPT partner can be defined. The minimal constituent chiral Yukawa operator is Fredholm at the essential-spectrum level but has trivial ordinary Callias boundary class. Its four-dimensional spectral flow is logically distinct from a static kernel. A universal degree-five differential character and its degree-two family transgression are correctly typed, but restricted gauge-stack descent is not proved. In particular, tested constant loops and the faithful centre do not exhaust the gauge group. Therefore*

$$\texttt{same\_physical\_moduli\_space\_derived=false}, \qquad \texttt{determinant\_line\_o2\_derived=false},$$
$$\texttt{gauge\_basic\_wzw\_descent\_proven=false}, \qquad \texttt{lane\_closed=false}.$$

The narrower distinction is

$$\texttt{restricted\_higgsed\_sigma\_model\_descent\_proven=false},$$
$$\texttt{restricted\_local\_differential\_character\_ansatz\_typed=true}.$$

Recent Skyrmion–fermion computations explicitly find a level that flows through zero as a Yukawa parameter is varied; they do not turn every static endpoint into a protected zero mode [1]. Recent generalized Callias theory likewise emphasizes that the index is a pairing of the actual potential class with the Dirac class and can be localized to an appropriate hypersurface [2]. These results motivate the separation enforced below.

2 The actual B=1 moduli: stabilizer before quantization

2.1 Relaxed hedgehog and its baryon number

Workline-B solves the coupled nonlinear (F, s) boundary-value problem and publishes a canonical sampling with box radius (R=16) and (4097) points. Its pion component defines

$$U_F(\boldsymbol{x}) = \cos F(r) + \mathrm{i}\tau_i \widehat{x}_i \sin F(r), \qquad F(0) = \pi, \quad F(\infty) = 0. \tag{1}$$

With the convention

$$B = -\frac{1}{24\pi^2} \int_{\mathbb{R}^3} \mathrm{Tr}\big(U_F^{-1} \mathrm{d}U_F\big)^3, \tag{2}$$

the canonical numerical solution gives $B = 0.999999999686391$ and $s(0) = 0.917128964607058$. Neither topology nor spherical symmetry identifies the moduli space with S^2.

The verifier calls exactly

```
scan_ap_e6_relaxed_b1_multigrid_hessian
.solve_relaxed_profile(box=16,sample_points=4097)
```

and forms the SHA-256 digest of the little-endian float64 columns (r, F, s). It hard-asserts equality with the digest stored in the Workline-B machine card:

`81104f59a5f3f4337739fc2a217cafc8da91581c9f1fb40b4fdba6ce0f948d59`.

The full Workline-B script and card hashes are recorded in Table 1 and in the JSON source manifest. Thus "same soliton" means byte-identical canonical (r, F, s) data, not an independent AP-E3 solve.

2.2 Pure internal stabilizer

Let $A \in SU(2)_V$ act by $U_F(\boldsymbol{x}) \mapsto AU_F(\boldsymbol{x})A^{-1}$. Choose a radius r_0 for which $\sin F(r_0) \neq 0$, which exists for every nontrivial degree-one profile. If A fixes the complete configuration, then

$$A(\tau_i \widehat{x}_i)A^{-1} = \tau_i \widehat{x}_i \quad \text{for all } \widehat{\boldsymbol{x}} \in S^2. \tag{3}$$

Taking successively three independent directions implies $A\tau_i A^{-1} = \tau_i$ for $i = 1, 2, 3$. By irreducibility of the defining $SU(2)$ representation, or directly by the Pauli commutators,

$$\mathrm{Stab}_{SU(2)_V}(U_F) = \{\mathbf{1}, -\mathbf{1}\} = \mathbb{Z}_2. \tag{4}$$

Thus the pure internal orbit is

$$\mathcal{M}_{\text{int}} = SU(2)_V/\mathbb{Z}_2 \simeq SO(3), \qquad \dim_{\mathbb{R}} \mathcal{M}_{\text{int}} = 3. \tag{5}$$

The verifier forms the positive Gram matrix

$$G_{ab} = \sum_{i=1}^{3} \mathrm{Tr}\big([\tau_a, \tau_i]^\dagger [\tau_b, \tau_i]\big), \tag{6}$$

whose kernel is the continuous stabilizer Lie algebra. All three eigenvalues are strictly positive.

2.3 Combined spatial–isospin symmetry is the same orbit

It is essential not to count another three modes. Let $(R, A) \in SO(3)_{\text{space}} \times SU(2)_V$ act by

$$U_F(\boldsymbol{x}) \longmapsto AU_F(R^{-1}\boldsymbol{x})A^{-1}. \tag{7}$$

Writing $\pi : SU(2) \to SO(3)$ for the double cover, Eq. (1) is fixed precisely when $R = \pi(A)$. Hence the connected stabilizer is the diagonal lift $SU(2)_{\text{diag}}$, with the appropriate central identifications, and

$$\frac{SO(3)_{\text{space}} \times SO(3)_V}{SO(3)_{\text{diag}}} \simeq SO(3). \tag{8}$$

Equations (5) and (8) describe the same relative spin–isospin orientation of a hedgehog. They are not two independent $SO(3)$'s. Moreover, U_F is a colour singlet. Microscopic $SU(2)_c$ gauge transformations do not quotient this global pion orbit.

Corollary 2.1 (No physical $\mathbb{CP}^1$ from hedgehog rotations)**.** *The relaxed hedgehog has no $U(1)$ pure-internal stabilizer. Therefore the identification $SU(2)/U(1) = \mathbb{CP}^1$ cannot be made for this solution.*

2.4 Why the scalar vacuum sphere is not a soliton modulus

The charged UV candidate also has a scalar doublet $\phi = vz$, $z^\dagger z = 1$, $z \sim e^{\mathrm{i}\lambda} z$, so the vacuum quotient is $\mathbb{CP}^1$. If $z = z(m)$ is spatially uniform, the kinetic norm of a variation is

$$\|\delta_m \phi\|^2 = v^2 \int_{\mathbb{R}^3} \mathrm{d}^3x\, \delta_m z^\dagger (1 - zz^\dagger)\delta_m z. \tag{9}$$

For any nonzero tangent variation the density approaches a positive constant, so the norm diverges as the volume. In a box $[-L, L]^3$, $\|\delta_m \phi\|^2 \propto 8L^3$. This is a bulk Goldstone coordinate, not a localized collective coordinate. Making it slowly time-dependent produces an inertia that diverges in the infinite-volume limit.

Fail-closed gate 2.2 (Same physical moduli)**.** One would have to find a different relaxed charged soliton with a localized scalar texture and stabilizer $U(1)$, then prove its horizontal mode is L^2-normalizable. The present solution proves instead $\mathcal{M}_1 \simeq SO(3)$ for rotations and a nonnormalizable bulk $\mathbb{CP}^1$.

3 A concrete constituent Yukawa completion on the same profile

3.1 Four-dimensional Euclidean operator

The lowest-dimension chiral constituent coupling compatible with vectorlike $U(1)_g$ charge is declared to be

$$S_{F,Y} = \int \mathrm{d}^4x\, \overline{\Psi} \Big[\gamma^\mu(\partial_\mu + \mathcal{A}_\mu) + M\big(P_R U_F + P_L U_F^\dagger\big)\Big]\Psi, \qquad P_{R,L} = \frac{1 \pm \gamma_5}{2}. \tag{10}$$

Here Ψ carries spin and flavour/isospin; $\mathcal{A}_\mu$ denotes the declared vectorlike gauge connection. Equation (10) is a standard chiral Yukawa form, but in this project it is a *declared constituent completion*: the current Workline-B charged EFT supplied no fermion representation, Yukawa coefficient, or regulator matching. It must not be silently promoted to a renormalizable microscopic interaction.

The canonical Workline-B background also contains the relaxed breathing field $s(r)$. Equation (10) deliberately declares no $s\overline{\Psi}\Psi$, sU_F, or other breathing-scalar Yukawa coupling. Therefore s is *decoupled by this declared completion*; it is not silently omitted from the imported solution. Any microscopic matching that generates an s-dependent fermion mass changes the operator and requires a new Callias audit.

Under the Euclidean CPT image

$$U_F(\boldsymbol{x}) \mapsto U_F^\dagger(-\boldsymbol{x}), \qquad \mathcal{A}_\mu(x) \mapsto \mathcal{A}_\mu^{\mathrm{CPT}}(x), \tag{11}$$

assume an antiunitary map Θ on the one-particle Hilbert space such that $\Theta \mathrm{i} \Theta^{-1} = -\mathrm{i}$ and

$$\mathcal{D}_E[\Phi^{\mathrm{CPT}}] = \Theta\, \mathcal{D}_E[\Phi]\Theta^{-1}. \tag{12}$$

This relation is conditional: Θ must map the operator domain, boundary conditions, and spectral cut to their CPT images; the fermion and regulator representations must be Θ-closed; regulator masses must be real. It implies determinant conjugation only after those assumptions and a consistent determinant-line orientation are supplied. It does not choose a physical regulator.

3.2 Static three-dimensional Hamiltonian

Choose Pauli matrices ρ_i for the upper/lower Dirac block, σ_i for spin, and τ_i for isospin. One convenient Hermitian representation is

$$\alpha_i = \rho_1 \otimes \sigma_i, \qquad \beta = \rho_3 \otimes \mathbf{1}, \qquad \gamma_5 = \rho_1 \otimes \mathbf{1}. \tag{13}$$

On $\mathcal{H} = L^2(\mathbb{R}^3, S \otimes \mathbb{C}^2_\tau)$, define

$$D_3 = -\mathrm{i}\alpha^i D_i, \qquad \Phi_F = \beta M\big(\cos F + \mathrm{i}\gamma_5 \boldsymbol{\tau}\cdot\widehat{\boldsymbol{x}} \sin F\big), \qquad H_F = D_3 + \Phi_F, \tag{14}$$

initially on C_c^∞ and with self-adjoint domain $\operatorname{Dom} H_F = H^1$ under the smooth-decaying gauge assumptions below. Explicitly,

$$\begin{aligned} H_F = {} & -\mathrm{i}\rho_1 \otimes \sigma_i D_i \otimes \mathbf{1}_\tau + M\rho_3 \otimes \mathbf{1} \otimes \mathbf{1} \cos F(r) \\ & - M\rho_2 \otimes \mathbf{1} \otimes \big(\boldsymbol{\tau}\cdot\widehat{\boldsymbol{x}}\big) \sin F(r). \end{aligned} \tag{15}$$

The second line follows from $\mathrm{i}\beta\gamma_5 = -\rho_2$. Thus Φ_F and H_F are Hermitian. Notice also that $\Phi_F^2 = M^2\mathbf{1}$.

Since $F(r) \to 0$ exponentially in the Workline-B massive-pion profile,

$$H_F - H_\infty \text{ is relatively compact}, \qquad H_\infty = -\mathrm{i}\alpha^i\partial_i + \beta M. \tag{16}$$

Consequently

$$\sigma_{\mathrm{ess}}(H_F) = (-\infty, -|M|] \cup [|M|, \infty). \tag{17}$$

This proves a uniform *essential-spectrum* gap $|M|$ under global unitary isorotation. It does not exclude discrete eigenvalues in the gap, prove $\inf |\sigma(H_F)| > 0$, or establish a uniform Fredholm family over an unconstructed physical moduli space.

3.3 Ordinary Callias boundary index is zero

The self-adjoint Hamiltonian H_F is not itself the off-diagonal Fredholm operator whose index is being quoted. Define

$$\mathscr{C}_F = D_3 + \mathrm{i}\Phi_F : H^1(\mathbb{R}^3, S \otimes \mathbb{C}^2_\tau) \longrightarrow \mathcal{H}, \qquad \mathbb{B}_F = \begin{pmatrix} 0 & \mathscr{C}_F^\dagger \\ \mathscr{C}_F & 0 \end{pmatrix}, \tag{18}$$

where $\mathbb{B}_F$ acts on the doubled Hilbert space with grading $\Gamma = \mathrm{diag}(1, -1)$. We impose the following sufficient Callias hypotheses:

(C1) $\mathbb{R}^3$ is complete and D_3 is self-adjoint on H^1;

(C2) Φ_F is bounded and Hermitian, and $[D_3, \Phi_F]$ extends to a bounded operator;

(C3) outside a compact set K, for some $\epsilon > 0$,

$$\Phi_F^2 - \|[D_3, \Phi_F]\|\mathbf{1} \geq \epsilon\mathbf{1}; \tag{19}$$

(C4) the gauge field is smooth and decays sufficiently that $H_F - H_\infty$ is relatively compact.

Then $\mathscr{C}_F$ is Fredholm, $\mathbb{B}_F$ is its graded self-adjoint double, and the index is determined by the positive eigenbundle $E_+(\Phi_\infty) \to S^2_\infty$ [3, 2]. For Eq. (14), the correct boundary potential is

$$\Phi_\infty = \beta M, \qquad E_+(\Phi_\infty) = S^2_\infty \times \mathbb{C}^4, \qquad c_1(E_+) = 0, \qquad \mathrm{Ind}\,\mathscr{C}_F = 0. \tag{20}$$

The rank-four value refers to the displayed Dirac–isospin representation; only its constancy is topologically relevant. Equation (20) is asserted only under (C1)–(C4), not for an arbitrary microscopic gauge background. The fact that $U_F : \mathbb{R}^3 \cup \{\infty\} \simeq S^3 \to SU(2) \simeq S^3$ has degree one is a bulk π_3 statement. The ordinary static Callias boundary theorem sees S^2_∞ and a constant mass here; it cannot be replaced by B without changing the operator problem.

3.4 Where B does enter: four-dimensional spectral flow

Let U_τ interpolate from the vacuum to U_F, and define

$$\mathscr{D}_{\text{flow}} = \partial_\tau + H[U_\tau]. \tag{21}$$

With APS boundary conditions, a maintained endpoint gap, and a regulator whose anomaly normalization is fixed, the standard index/spectral-flow identity is

$$\mathrm{Ind}\,\mathscr{D}_{\text{flow}} = \mathrm{SF}\{H[U_\tau]\}. \tag{22}$$

In the adiabatic chiral problem the anomaly calculation identifies this spectral flow with the winding B, up to the stated representation and sign conventions [4, 5]. But spectral flow counts crossings *along the path*. It does not assert $\mathrm{Ker}\,H[U_F] \neq 0$. Numerically observed Skyrmion levels indeed cross zero only at selected coupling values [1].

Fail-closed gate 3.1 (Static Callias versus interpolation index)**.** For the same canonical background, the defensible statements are

uniform essential gap true,

ordinary boundary Callias index 0,

conditional interpolation spectral flow B.

The full endpoint spectral gap and a physical static zero-mode bundle remain unproved.

4 Families determinant line and a two-band obstruction

4.1 Conditional push-forward formula

To compare with the previous template, suppose temporarily that a physical $\mathbb{CP}^1$ family Y existed and a rank-one positive asymptotic eigenbundle over

$$X = S^2_\infty \times Y, \qquad H^*(X;\mathbb{Z}) = \mathbb{Z}[x,y]/(x^2,y^2), \qquad \int_{S^2_\infty} x = 1 \tag{23}$$

had

$$c_1(E_+) = px + qy. \tag{24}$$

Since

$$\mathrm{ch}(E_+) = 1 + px + qy + pq\,xy, \tag{25}$$

the oriented boundary push-forward gives

$$\mathrm{rank}\,\mathrm{Ind}\,\mathscr{C} = \epsilon p, \qquad c_1(\det \mathrm{Ind}\,\mathscr{C}) = \epsilon pq\,y. \tag{26}$$

Thus $\epsilon p = 1, q = 2$ would produce a rank-one $\mathcal{O}(2)$ index line. The Bismut–Freed determinant connection would then refine Eq. (26) geometrically [6, 7].

4.2 No globally gapped two-band realization of x+2y

Theorem 4.1 (Two-band mixed-line obstruction)**.** *Let $\Phi : X \to \mathrm{Herm}(2)$ be continuous on $X = S^2 \times \mathbb{CP}^1$, acting on the trivial bundle $X \times \mathbb{C}^2$. Assume that its two eigenvalues are globally nondegenerate and separated by zero, with exactly one positive band. Then its positive eigenline E_+ obeys*

$$c_1(E_+)^2 = 0. \tag{27}$$

In particular $c_1(E_+) = x + 2y$ is impossible.

Proof. Subtracting half the trace and dividing by the nonzero eigenvalue separation deformation-retracts Φ to a traceless unit Hermitian matrix

$$\widehat{\Phi} = f_a \tau_a, \qquad \boldsymbol{f} : X \longrightarrow S^2 \simeq \mathbb{CP}^1. \tag{28}$$

The positive eigenline is $E_+ = f^*\mathsf{H}_+$, where $\mathsf{H}_+ \to \mathbb{CP}^1$ is the universal positive eigenline. If $u = c_1(\mathsf{H}_+)$, then $u^2 = 0$ because $H^4(\mathbb{CP}^1;\mathbb{Z}) = 0$. Naturality gives

$$c_1(E_+)^2 = f^*(u^2) = 0.$$

On the other hand, using $x^2 = y^2 = 0$,

$$(x + 2y)^2 = 4xy, \tag{29}$$

and xy generates $H^4(S^2 \times \mathbb{CP}^1;\mathbb{Z})$. Hence $4xy \neq 0$, a contradiction. □

The hypotheses matter. The theorem does not cover a nontrivial ambient rank-two bundle, a band crossing, more than two bands, or an operator-valued infinite-dimensional mass. It does prove that the most economical globally gapped 2×2 spectator mass cannot realize the desired mixed line. At least a rank-three ambient construction or nontrivial microscopic bundle is necessary, but not by itself sufficient.

4.3 Why the Veronese control is not the missing construction

For $z = (z_0, z_1) \in \mathbb{C}^2$, define

$$u_2(z) = \left(z_0^2, \sqrt{2} z_0 z_1, z_1^2\right)^T, \qquad K(z) = \mathbf{1}_3 - u_2(z) u_2(z)^\dagger. \tag{30}$$

Then $K u_2 = 0$, the two nonzero eigenvalues equal one, and $u_2(e^{\mathrm{i}\lambda} z) = e^{2\mathrm{i}\lambda} u_2(z)$. The kernel line over $\mathbb{CP}^1$ is $\mathcal{O}(2)$. The verifier computes its Berry number $+2$ using gauge-invariant Bargmann triangle phases.

This is a rank-three *spectator* control over the y-sphere only. It does not construct the required $p = 1$ dependence over S^2_∞, does not couple to Eq. (15), and is not a Yukawa term of the charged mother model.

4.4 The actual orientation space cannot carry free c1=2

Even before choosing a mass, the physical orbit found above has

$$H^2(SO(3); \mathbb{Z}) \simeq \mathbb{Z}_2, \qquad H^2(SU(2); \mathbb{Z}) = H^2(S^3; \mathbb{Z}) = 0. \tag{31}$$

Thus an orientation determinant line can at most carry a torsion $\mathbb{Z}_2$ class on $SO(3)$; on the spin cover it is topologically trivial. There is no free Chern number $+2$ to identify with $\mathcal{O}(2)$. A possible torsion sign is a Finkelstein–Rubinstein/fermion-measure question and is not computed here.

Fail-closed gate 4.2 (Families determinant)**.** The abstract formula (26) and the Veronese regression are correct. The same soliton supplies neither their $\mathbb{CP}^1$ base nor the mixed asymptotic eigenbundle. Therefore `determinant_line_o2_derived=false`.

5 CPT algebra, Pfaffians, and the conditional anti-canonical map

5.1 Pauli–Villars determinant transformation

Let $\mathcal{D}_M[\Phi]$ be Eq. (10) and choose a CPT-covariant Pauli–Villars operator $\mathcal{D}_\Lambda[\Phi]$ with the same kinetic representation. Define

$$Z_{\mathrm{PV}}[\Phi] = \frac{\det \mathcal{D}_M[\Phi]}{\det \mathcal{D}_\Lambda[\Phi]}. \tag{32}$$

The following are assumptions, not outputs of the charged EFT:

(T1) Θ is antiunitary, obeys $\Theta \mathrm{i} \Theta^{-1} = -\mathrm{i}$, and satisfies Eq. (12);

(T2) Θ maps the operator domain, boundary conditions, and spectral cut to their CPT images;

(T3) the physical and Pauli–Villars representations are closed under Θ, and every regulator mass is real;

(T4) determinant and Pfaffian orientations are transported consistently.

Under (T1)–(T4), for both masses,

$$Z_{\mathrm{PV}}[\Phi^{\mathrm{CPT}}] = Z_{\mathrm{PV}}[\Phi]^*. \tag{33}$$

Equivalently, its eta/phase contribution reverses sign. A finite spectral cutoff regression uses real eigenvalues λ_j and

$$Z_+(\mu, \Lambda) = \prod_j \frac{\mu + \mathrm{i}\lambda_j}{\Lambda + \mathrm{i}\lambda_j}, \qquad Z_- = \prod_j \frac{\mu - \mathrm{i}\lambda_j}{\Lambda - \mathrm{i}\lambda_j} = Z_+^*. \tag{34}$$

This is an algebraic finite spectral-product identity, not a microscopic regulator construction. Global phases of fermion determinants require eta/Dai–Freed data beyond a local matrix identity [8].

5.2 Majorana doubling and Pfaffian orientation

For an antisymmetric doubled matrix

$$\mathcal{A}(D) = \begin{pmatrix} 0 & D \\ -D^T & 0 \end{pmatrix}, \qquad \operatorname{Pf}\mathcal{A}(D) = (-1)^{n(n-1)/2} \det D, \tag{35}$$

Eq. (33) implies the corresponding Pfaffian conjugation after a fixed orientation. Whether the orientation can be fixed globally on all gauge sectors is precisely an anomaly/regulator question and remains open.

5.3 Anti-canonical O(-4) remains conditional

If a physical polarized $\mathbb{CP}^1$ family with $E_+ = \mathcal{O}(2)$ existed, fixed Dolbeault polarization and Serre duality would map it to

$$E_- = K_{\mathbb{CP}^1} \otimes E_+^\vee = \mathcal{O}(-2) \otimes \mathcal{O}(-2) = \mathcal{O}(-4). \tag{36}$$

Then $(h^0, h^1) = (3, 0)$ maps to $(0, 3)$. But Eq. (36) is a statement on $\mathbb{CP}^1$. The actual orientation space is $SO(3)$, and the mother model has not supplied the polarization or the anti-canonical regulator factor. Determinant conjugation alone does not manufacture them.

Fail-closed gate 5.1 (Physical CPT regulator)**.** The determinant/Pfaffian conjugation law follows conditionally from (T1)–(T4), and its finite spectral-product regression is algebraically correct. No Θ-covariant regulator has been derived from the mother model, and the $\mathcal{O}(-4)$ physical line is absent. Hence `physical_cpt_regulator_derived=false`.

6 Typed local mixed-WZW character and the unclosed descent problem

6.1 Universal character before evaluation

Restrict first to $\phi^\dagger\phi = v^2 > 0$ everywhere. Write $z = \phi/v \in S^3$, with $z \mapsto e^{\mathrm{i}\lambda} z$, and let

$$s = [z] : M_4 \longrightarrow \mathbb{CP}^1. \tag{37}$$

Let $\mathsf{H} \to \mathbb{CP}^1$ be the Hopf line and $\widehat{c}_1(\mathsf{H}) \in \widehat{H}^2(\mathbb{CP}^1; \mathbb{Z})$ its differential first Chern class. Let $\widehat{\omega}_3 \in \widehat{H}^3(SU(2); \mathbb{Z})$ have curvature the integrally normalized baryon three-form. The correctly typed universal object is defined on the target product, before any spacetime field is inserted:

$$\boxed{\widehat{\Xi}_5 = 2\, \mathrm{pr}_U^* \,\widehat{\omega}_3 \smile \mathrm{pr}_s^* \,\widehat{c}_1(\mathsf{H}) \in \widehat{H}^5(SU(2) \times \mathbb{CP}^1; \mathbb{Z}).} \tag{38}$$

This is an intrinsic differential-cohomology construction, not a choice of a five-dimensional extension [9]. The coefficient two is the charged-two-colour mixed-WZW level matched in the intermediate UV candidate [10]; global-form refinements of symplectic QCD remain relevant [11].

Let $\mathcal{C}_{\mathrm{restr}}$ denote the smooth nonvanishing-Higgs field space before quotienting by the gauge group. Its evaluation map is

$$\mathrm{ev} : \Sigma_3 \times \mathcal{C}_{\mathrm{restr}} \longrightarrow SU(2) \times \mathbb{CP}^1, \qquad (x, c) \longmapsto \big(U_c(x), s_c(x)\big). \tag{39}$$

Only now do we pull back and fiber-integrate:

$$\widehat{\mathcal{L}}_{\mathrm{loc}} := \int_{\Sigma_3} \mathrm{ev}^* \,\widehat{\Xi}_5 \in \widehat{H}^2(\mathcal{C}_{\mathrm{restr}}; \mathbb{Z}). \tag{40}$$

Thus $\widehat{\mathcal{L}}_{\mathrm{loc}}$ is a well-typed line-with-connection candidate on the unquotiented restricted field space. Equation (40) alone does not construct an equivariant differential cocycle or prove that the line descends to the gauge quotient stack.

6.2 Local curvature control is not stack descent

On a local Hopf lift define

$$a = -\mathrm{i}z^{\dagger}\mathrm{d}z, \qquad z \mapsto e^{\mathrm{i}\lambda}z, \qquad a \mapsto a + \mathrm{d}\lambda. \tag{41}$$

The curvature can be written without the lift,

$$f = \mathrm{d}a = -\mathrm{i}\,\mathrm{d}z^{\dagger}(1 - zz^{\dagger}) \wedge \mathrm{d}z = s^{*}F_{\mathrm{FS}}. \tag{42}$$

It is invariant and horizontal under the vertical $U(1)$ vector. Since U is neutral under vector $U(1)_g$, the curvature $2\omega_3(U) \wedge f/(2\pi)$ passes the local horizontal-curvature test. The connection a itself is not a basic one-form; rather, its transformation law is exactly the descent datum of the associated line bundle. Confusing these two statements would incorrectly demand a globally defined potential on $\mathbb{CP}^1$. Conversely, horizontal curvature does not by itself prove equivariant differential-character descent: one must still specify the cocycle on gauge arrows and prove trivial vertical holonomy.

6.3 A sampled constant-loop integrality control

For the restricted constant-loop ansatz and a gauge function with winding $\lambda(1) - \lambda(0) = 2\pi n$, the worldline connection at baryon number B shifts by

$$\Delta S/\hbar = 2B\,[\lambda(1) - \lambda(0)] = 4\pi Bn. \tag{43}$$

Therefore

$$\exp(\mathrm{i}\Delta S/\hbar) = 1 \qquad (B, n \in \mathbb{Z}). \tag{44}$$

The quantized coefficient is essential. This computation samples a one-parameter class; it does not classify all connected components of $\mathrm{Map}(\Sigma_3, U(1))$, nor does it supply their equivariant cocycle data.

6.4 Free Hopf action and faithful centre

The $U(1)$ action on S^3 is free whenever $\phi \neq 0$, so the restricted Hopf quotient has no gauge stabilizer. Separately, the faithful global identification equates the flavour centre $z \mapsto -z$ with the gauge transformation $e^{\mathrm{i}\pi}$. An $\mathcal{O}(k)$ fiber transforms by

$$e^{\mathrm{i}k\pi} = (-1)^k. \tag{45}$$

Thus $k = 2B$ has trivial phase for this tested centre element, while the $k = 1$ negative control does not. This is a necessary even-weight check. It is *not* an exactly-two theorem: it neither excludes higher even weights nor checks every stabilizer, gauge component, global form, or anomaly constraint.

6.5 Restriction to the two candidate collective spaces

If one formally inserts a product family $(U_B(\boldsymbol{x}), s(m))$ with $m \in \mathbb{CP}^1$, then

$$c_1(\widehat{\mathcal{L}}_B) = \int_{\Sigma_3 \times \mathbb{CP}^1} 2\,\omega_3(U_B) \wedge s^{*}\omega_2 = 2B\deg(s). \tag{46}$$

For $B = 1, \deg(s) = 1$, this is $+2$. But this $s(m)$ is the volume-divergent scalar vacuum coordinate of Eq. (9). It is not a localized moduli-space map.

On the actual rotational orbit $SO(3)$, hold the scalar vacuum fixed. The evaluation map has no scalar tangent component, so the two-form curvature of Eq. (40) restricts trivially along the $SO(3)$ orbit. The WZW line therefore does not supply a free $c_1 = 2$ on the physical orientation family.

6.6 Why even restricted stack descent remains open

Equations (38)–(40) are correctly typed on the unquotiented nonvanishing-Higgs field space. To descend them to the restricted gauge stack one must still:

(R1) treat every component of $\mathrm{Map}(\Sigma_3, U(1))$, not only the sampled constant-loop ansatz;

(R2) construct the equivariant differential cocycle on gauge arrows and verify its composition law;

(R3) prove trivial vertical holonomy for every gauge loop.

The extension to the full microscopic space additionally requires:

(D1) configurations where $\phi = 0$ and the projective map s fails;

(D2) defects and nontrivial gauge bundles not represented by one global normalized doublet;

(D3) the full faithful global-form quotient and all non-extendible bordism sectors of the microscopic theory;

(D4) a same-soliton normalizable collective map.

Accordingly the machine card distinguishes the two flags

```
restricted_local_differential_character_ansatz_typed=true
restricted_higgsed_sigma_model_descent_proven=false
gauge_basic_wzw_descent_proven=false
```

7 Numerical controls

7.1 Same-profile and stabilizer regressions

The verifier imports the Workline-B public solver, calls it at $R = 16$ with (4097) samples, and hard-asserts equality of the canonical (r, F, s) checksum with the Workline-B card. It then reads B from that same card, evaluates the Gram matrix (6), and fits the uniform scalar-mode norm against box size. The expected power is three.

7.2 Finite Wilson–Dirac endpoint control

On a cubic grid the verifier discretizes Eq. (15) with central derivatives and a Wilson control term

$$H_{W,a} = H_{F,a} - \frac{r_W a}{2}\,\beta\Delta_a. \tag{47}$$

The mass term uses only the canonical Workline-B (F) samples, with a stated piecewise-linear off-grid interpolation. The accompanying canonical breathing field (s) does not enter because the declared coupling (10) contains no (s)-dependent Yukawa term. Dirichlet finite-box boundaries are used. The calculation checks Hermiticity and the smallest-magnitude eigenvalues on two grids. A positive finite-box minimum excludes an accidental zero only for this benchmark. It is not a continuum extrapolation, a fermion determinant, or lattice QC2D. The grids are deliberately small and the solver uses shift-invert `eigsh` with a smallest-magnitude fallback for reproducibility.

There is also a direct coset-representative test. If one incorrectly parametrizes the hedgehog by $SU(2)/U(1)$, then A and $Ae^{-i\theta\tau_3/2}$ denote the same proposed $\mathbb{CP}^1$ point. The two conjugated hedgehog operators differ at nonzero θ, although they are unitarily isospectral. Hence the operator does not descend to that $\mathbb{CP}^1$ quotient.

7.3 Berry, CPT, and holonomy controls

The following controls are deliberately finite and diagnostic:

(N1) Bargmann-triangle Berry phases of Eq. (30) give $c_1 = +2$, with unit nonzero projector gap.

(N2) The spectral product (34) and its antisymmetric doubling obey determinant and Pfaffian conjugation to floating-point precision.

(N3) Sampled constant-loop $U(1)$ windings give unit phase at level two; the faithful centre gives $+1$ at level two and -1 in the level-one control. These do not prove restricted gauge-stack descent.

7.4 Recorded numerical values and scope

The profile and its producing artifacts are cryptographically bound as follows.

Artifact	SHA-256
Canonical little-endian float64 (r, F, s)	`81104f59a5f3f4337739fc2a217cafc8da91581c9f1fb40b4fdba6ce0f948d59`
Workline-B solver script	`393a1a1948b65bceb3c04167e452140a643a482a5035ed419d892fdd9cc1f10d`
Workline-B machine card	`204dc3c91d3b83031fb70cafefa63e29df56f14c76c230600497ec94e1fb18b5`

Table 1: Canonical same-soliton provenance used by the C workline.

Table 2 records the deterministic run. The two Wilson entries are deliberately reported separately: their increase with resolution is not presented as a continuum extrapolation. They only show that neither finite matrix has an accidental zero at the declared benchmark.

Table 2: AP-E6 deterministic numerical controls.

Quantity	Value	Interpretation
Canonical Workline-B baryon number	0.999999999686391	Card-bound $R = 16$ coupled profile.
Canonical breathing field at the origin	$s(0) =$ 0.917128964607058	Imported and explicitly decoupled by the declared Yukawa.
Stabilizer Gram spectrum	$(16, 16, 16)$	No continuous pure-internal stabilizer.
Uniform scalar norm power	3.000000000000001	Volume divergence in a cubic box.
Wilson 5^3, matrix dimension 1000	$\min\lvert\lambda\rvert =$ 0.010896961	Finite-box endpoint control only.
Wilson 7^3, matrix dimension 2744	$\min\lvert\lambda\rvert =$ 0.283674971	Finite-box endpoint control only.
False-$\mathbb{CP}^1$ representative operator difference	1.15817×10^{-2}	Relative Frobenius norm.
Representative isospectral residual	4.86×10^{-17}	Unitary-conjugation check.
Veronese spectator Berry number	1.999999999999986	$\mathcal{O}(2)$ template, not same-model evidence.
Veronese kernel residual / nonzero gap	$4.16 \times 10^{-16}/1$	Projector regression.
CPT determinant / Pfaffian residual	0/0	Finite spectral-cutoff algebra.

Quantity	Value	Interpretation
Maximum level-two large-gauge phase residual	2.45×10^{-15}	Sampled windings only; not stack descent.

Fail-closed gate 7.1 (Meaning of a green verifier)**.** A green run means that the no-go deductions, conditional index formulae, finite discretizations, and fail-closed flags are mutually consistent. It does not mean that a physical Callias family or portal has been constructed.

8 Gate ledger and next mathematically decisive experiments

Gate	Status	Reason
Same physical $\mathbb{CP}^1$ moduli space derived	false	Hedgehog orbit is $SO(3)$; scalar $\mathbb{CP}^1$ norm diverges with volume.
Actual Yukawa–Callias operator derived from mother model	false	Eq. (10) is a declared constituent completion.
Uniform essential Fredholm gap	true	Under (C1)–(C4), $\Phi_\infty = \beta M$ gives essential threshold $\lvert M \rvert$.
Uniform full endpoint spectral gap	false	Only two finite Wilson controls; no continuum exclusion of zero crossings.
Static ordinary Callias index +1	false	Actual asymptotic positive eigenbundle is trivial and has index zero.
Four-dimensional interpolation formula	conditional theorem	APS index equals spectral flow; anomaly matching gives B under stated regulator assumptions.
Same-model determinant line $\mathcal{O}(2)$	false	Wrong base and no mixed mass; minimal globally gapped two-band mass is obstructed.
Physical CPT/anti-canonical regulator	false	Conditional algebra holds under (T1)–(T4), but no physical regulator or $\mathbb{CP}^1$ polarization is derived.
Restricted local differential-character ansatz typed	true	Universal target character, evaluation pullback, and fiber-integration degrees are consistent.
Restricted Higgsed gauge-stack descent	false	All gauge-group components, an equivariant cocycle, and vertical holonomy remain unchecked.
Full microscopic gauge-basic descent	false	Higgs zeros, defects, all bundles/bordism sectors, and same-soliton pullback remain open.
Same-soliton composition / lane closed	false	No physical $\mathbb{CP}^1$, Callias $\mathcal{O}(2)$, or CPT line.
Degree-one Route-E portal	not authorized	No pre-portal lane is closed.

There are three concrete ways forward, ordered by how directly they address the no-go.

C1. Change the soliton, not its name. Search the full relaxed gauge–meson–scalar equations for a localized scalar texture whose exact symmetry breaking is $G/H = SU(2)/U(1)$, and verify the horizontal L^2 norm and projected Hessian gap. This is the only route that keeps the desired physical $\mathbb{CP}^1$ literal.

C2. Accept $SO(3)$ and reformulate the target. Compute the torsion determinant line, Finkelstein–Rubinstein sign, and WZW holonomy on $SO(3)$ or its $SU(2)$ cover. This may yield a meaningful two-valued structure, but it cannot reproduce a free Chern number two.

C3. Construct a genuine higher-rank asymptotic mass. Starting from anomaly-free microscopic fields, build a globally gapped rank-three or larger mass over $S^2_\infty \times \mathbb{CP}^1$, compute its family index and heavy-regulator threshold, and show that its light kernel couples to the same soliton. The Veronese control supplies only the $q = 2$ factor; the $p = 1$ spatial factor and decoupling are still missing.

No degree-one portal should be started until one of these routes closes all of its physical gates.

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